

Project Guidelines

Launch Systems 2025-26

Delivery

The project will be delivered on

- **Friday, Dec. 19th at MIDNIGHT (11:59 PM CET), for groups that want to take examination in the first exam session (Jan/Feb 2026).**
- **Friday, May 22nd at MIDNIGHT (11:59 PM CEST), for groups that want to take examination in the second/third exam session (June/Sept 2026).**

A template for the report is available on line but its use is not mandatory. You should only keep the font size (11 pt) and the page size (A4).

Maximum number of pages is 30 (excluding front pages, indexes, appendix, and reference).

Delivery file: single PDF file max 10 MB (no extra files or codes, unless requested afterwards)

The delivery will be done through a link (Microsoft FORMS) where you should introduce the following data for EACH member: collect the data in advance.

- Complete name
- Personal ID code (10xxxxxx)
- Email

The student listed FIRST in the list of participants will act as a contact point for the whole group.

One submission for each group of people.

Remember that you can submit only ONCE. So, deliver the report in its final state.

We are expecting a report where you present your design, you show your performance, and you justify with numbers your approach. You have to document the models you are using (do not report the whole theory, only main equations and reference the rest).

It is not important if you describe the iterations of your work (unless they are mandatory to explain some important choice); it is more important that you present your final result, show the expected performance, and defend the method you have used. You have to convince a set of technical audience about the goodness of your results and that your choices are reasonable.

WARNING: an authorship declaration section is needed. You need to state who did what. On that basis, **we do questions and we select the person that will answer**. However, we expect that the **whole team** can reply to questions regarding **high level considerations** of the project and how **subsystems interrelate** each other (subsystem details are in charge of who developed them). This is a proof that project development was a real cooperative action. Questions will be done to clarify project text and integrate the presentation.

The authorship declaration is a responsibility of the team, and the capacity to reply to questions affects the evaluation of the whole team.

What you get

Launch systems FULL exam is composed by two written parts: THEORY and DESIGN. Both parts are done in the same day. Approx. duration 4 hours in total.

The project entitles all the authors to use this evaluation to skip the second part of the exam (DESIGN). This right is acquired once your project score (team + personal score) is sufficient 18 and it is valid up to the beginning of the next Launch Systems course.

Some rules/limitations apply. Please read carefully.

- If you do not deliver a project by December and you try the full exam in the first session (two normal classwork assignments), you lose the possibility to use the project to skip the second part of the exam. This limitations applies to the single student. The rest of the group is unaffected. *Rationale > you cannot try the exam in Jan/Feb if you are committed to deliver a project with a team by May.*
- Once delivered, you can opt if you want to take the exam with or without the project option.
- You can take the full exam without being part of a team.

Presentation

The teams will prepare a presentation. The team manages the presentation and appoints one or more representatives to present the work. Representatives must be in presence (unless specific reasons are given)

All team members are requested to be either on line or in presence because everyone may be asked to answer a question.

No format or constraint is posed on presentation style. The presentation slot is 10 minutes long. Project discussion will be given in the first week of January (slot reservation will open soon). The presentation must be given in presence and at least one member must be in class. The team can assist via WEBEX and can take questions via WEBEX as well, if needed. The presentation is public.

Scoring grid

Each team can obtain a maximum of 30 points. The following grid explains the evaluation methodology.

Report	Max	Presentation	Max
Structure	3	Clarity of presentation	4
Clarity of the text	4	Correct timing	2
Rigorous or not	4	Structure and order	3
Respect of requirements	4	Answers to questions	3
Original developments	3		

Here you can find the description of each item.

Report

- **Structure:** we evaluate the logical sequence of the matter, the presence and the format of figures, equations, tables, table of content, nomenclature, proper use of literature reference and if they are reported properly, cross references, presence of all report components (abstract, conclusions, ...), proper use of appendix.
- **Clarity:** we evaluate if the development process is clear, if we understand the method, if the results and problems are properly commented and discussed, if you used the management part (House of Quality properly commented) as a support to development of design requirements, if technical English is adequate.
- **Rigorous:** we evaluate if the applied methodology is correct, if assumptions are valid and reasonable, if validations to your models are present or properly justified/discussed, if you placed justified choices, if a minimal set of (correct) equations supporting your developments is reported, if you properly used a baseline.
- **Respect of requirements:** we evaluate if you developed properly requirements, and if you matched all of them or, in case of mismatch, if you explained the fact and you found a way-forward to address future development for improvement; these justifications and comments should be supported by numerical results available in your report.
- **Original developments:** we evaluate if you developed something that was not part of the course, independently or you extended the assignment (pay attention extension does not mean change ... come to us to agree on this part); the part should be supported by adequate reference list, modeling, and result analysis.

Presentation

- **Clarity:** we evaluate how you explain methodology, requirements, hypotheses, limitations, how you comment your results and how these have consequence on your final goal.
- **Timing:** you have in total 10 minutes; you need to stay within this time slot.
- **Structure and order:** we evaluate if your presentation has a reasonable structure and sequence, in proper logical order, if figures, tables and other graphical items are readable, if the global format is clear and the presentation software is not misused (e.g., confusing animations).
- **Answer to questions:** we evaluate if the question is properly answered, if the correct sense of the reply is well addressed, if the reply is reasonable and technically convincing.

We are planning an additional time for project score discussion, where the group can discuss the comments received in the project. This is still under development and will be before/after presentations.

Final score

The final score is composed by three parts:

- Project evaluation (30 points max)

- Peer evaluation (2 points max)
- Theory part (12 points max)

The final score is composed with the following process:

- The project activity and the peer evaluation will sum up. This will give the individual evaluation of the project activity (Max 30 + 2 = 32 point).
- If you do not use the project, your DESIGN part is scored 32 at most. Peer evaluation is neglected.
- The theory will give you a max of 12 points, with a sufficient level of 7 points. This limit is checked after the application of the bonus point. These total theory points will be scaled to 32 according to the following rationale:
 - 7 theory points (sufficient) corresponds to 18 in the theory score.
 - 12 points (excellence) corresponds to 32 in the theory score.
 - The conversion uses the following formula: $\text{Score} = \frac{(\text{Points} - 7)}{5} * 14 + 18$
- Project and theory scores are then averaged and rounded to the closest integer number (0.5 rounds upward)

Quick comment on new scoring

This new scoring mechanism has been implemented following some comments. After a series of “experiments”, the requirements leading to this scoring mechanisms are the following:

- Give more importance to the theory part without being too disruptive;
- Award the excellence, giving people the possibility to improve the project outcomes, if the theory part is well done (and vice versa).

Peer evaluation

Despite some of the the students are not in favor of peer evaluations, we think this is a good method to improve the positive collaboration inside groups. In your future career, you will be asked (sometimes forced) to work in teams and you will be subject to judgement (implicit or explicic) of your colleagues. We think that a honest use of peer evaluation is a good indication for everyone. We cannot exclude the malicious use of this instrument: every instrument can be used in an improper manner: there is no way to avoid that. I believe in fair thinking, and I am assuming that improper use of this instrument will not occur here. In principle, I cannot remove an important instrument only because there is a possible misuse in a strict minority of cases. I believe that you can understand the importance of a choice done by the teachers, if properly supported by explanation.

On teachers’ side we can improve the quality of peer evaluation by guiding your judgement though clear guidelines. In this present case, remember this:

- You are scoring the commitment, the capacity of collaboration, the correctness, the exchange capacity
- You ARE JUDGING NORMAL levels of performance and collaboration (2 points means that everything was simply fine)

What you are NOT judging:

- You are NOT scoring for excellence
- You are NOT scoring the fact that someone could not accomplish a task because he/she tried hard but he/she failed, if they interacted with other people to solve the issue and minimize the consequence. These people tried their best and acted properly so they deserve the maximum.
- You are scoring bad if someone did not collaborate appropriately, accepted a commitment but did not deliver the promised work on time on a regular basis, created some problems to the team because of lack of communication/collaboration/laziness etc.
- Remember that most of problems in teams (in general) are originated by lack of proper and clear communication. Solutions can always be found, if the problems are clearly stated and known in advance.

If you read up to this point, thank you for your attention.

Sorry for this long document.